Microeconomics exam

University, BSc Economics, year 1

Time: 120 minutes **Total:** 40 points

Answer all questions. Show your working for calculations. An answer without working will not earn full marks.

Question 1 (3 points)

On Saturday, Sanne can work 8 hours in a café for €14 per hour, or study for this exam. She has paid €2,530 in tuition fees and bought a textbook for €60.

a. Calculate the opportunity cost of spending Saturday studying.

b. Explain why the tuition fees and the textbook should play no role in her decision for Saturday.

Question 2 (3 points)

Consider the market for e-bikes.

a. Cheaper batteries cause the price of e-bikes to fall. Is this a movement along the demand curve or a shift of the demand curve? Explain.

b. The price of petrol rises sharply. What happens to the demand curve for e-bikes? Explain.

Question 3 (5 points)

In the market for bicycle lights (Q in thousands of units, P in euros):

Qd = 200 − 4P

Qs = −40 + 2P

a. Calculate the equilibrium price and quantity.

b. Calculate consumer surplus and producer surplus at the equilibrium.

Question 4 (5 points)

In the market from question 3, the government levies an excise tax of €6 per unit, paid by the producer.

a. Calculate the new equilibrium.

b. Calculate the tax revenue and the deadweight loss.

c. Who bears most of the tax, consumers or producers? Explain using elasticities.

Question 5 (4 points)

Take the equilibrium from question 3 (P = 40, Q = 40).

a. Calculate the price elasticity of demand at this point.

b. Is demand elastic or inelastic here, and what happens to total consumer spending if the price rises slightly?

Question 6 (4 points)

De Korenaar bakery has fixed costs of €12,000 per month, variable costs of €1.50 per loaf and a selling price of €3.50.

a. Calculate the break-even quantity.

b. At 8,000 loaves per month, calculate the profit and the average total cost.

Question 7 (5 points)

A firm holds a patent on a medical test and is the only supplier. The inverse demand function is P = 100 − 2Q. Marginal cost is constant: MC = 20. There are no fixed costs.

a. Determine the profit-maximising quantity and price.

b. Calculate the profit.

c. Calculate the deadweight loss compared with perfect competition.

Question 8 (3 points)

A chemical plant discharges waste water into a river, so fishermen downstream catch less.

a. Using marginal costs, explain why the market leads to an inefficient outcome.

b. Describe a Pigouvian tax: how high should it be, and why does it restore efficiency?

Question 9 (3 points)

Two airlines each choose a high or low price. Profits in millions of euros (A ; B):

	B high	**B low**
A high	8 ; 8	2 ; 10
A low	10 ; 2	4 ; 4

a. Does A have a dominant strategy? Determine the Nash equilibrium.

b. Explain why this game is a prisoner’s dilemma.

Question 10 (5 points)

Complete the table. The table has **5 columns**: number of firms, product (homogeneous or differentiated), barriers to entry, pricing (relation between P and MC) and long-run economic profit. The **4 rows** are the four market structures.

Market structure	Number of firms	Product	Barriers to entry	Pricing	Long-run economic profit
Perfect competition					
Monopolistic competition					
Oligopoly					
Monopoly					